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Problem Statement

In a convex quadrilateral $ABCD$, it holds that $|\angle ABC| = |\angle ADC|$ and $|\angle BCD| = 3|\angle BAD|$. Points P and Q lie on segments AB and AD respectively, such that $APCQ$ is a parallelogram. Let O be the circumcenter of triangle CPQ. Prove that $AO \perp BD$.

Proof

Step 1: Angle Analysis

Let $\alpha = |\angle BAD|$. Given that $|\angle BCD| = 3\alpha$ and the sum of angles in a quadrilateral is $360°$, we have:

$$\alpha + |\angle ABC| + 3\alpha + |\angle ADC| = 360°$$

Since $|\angle ABC| = |\angle ADC|$, let us denote their measure by β. The equation becomes:

$$4\alpha + 2\beta = 360° \implies 2\alpha + \beta = 180° \implies \beta = 180° - 2\alpha$$

Thus, angles $\angle ABC$ and $\angle ADC$ are supplementary to 2α.

Step 2: Properties of Triangles PBC and QDC

Since $APCQ$ is a parallelogram, we have $PC \parallel AD$ and $CQ \parallel AB$. Also, opposite angles of the parallelogram are equal: $|\angle PCQ| = |\angle PAQ| = \alpha$.

Consider $\triangle PBC$:

- The angle at vertex B is $\beta = 180° - 2\alpha$.
- Since $PC \parallel AD$ and AB intersects them, $\angle BPC$ corresponds to $\angle DAB$, so $|\angle BPC| = \alpha$.
- The remaining angle is $|\angle BCP| = 180° - (180° - 2\alpha) - \alpha = \alpha$.

Since $|\angle BCP| = |\angle BPC| = \alpha$, $\triangle PBC$ is isosceles with $\mathbf{PB} = \mathbf{BC}$.

Similarly for $\triangle QDC$:

- The angle at vertex D is $\beta = 180° - 2\alpha$.
- Since $CQ \parallel AB$ and AD intersects them, $|\angle DQC| = \alpha$.
- The remaining angle is $|\angle QCD| = \alpha$.

Thus, $\triangle QDC$ is isosceles with $\mathbf{QD} = \mathbf{DC}$.

Step 3: Characterizing Point O

Point O is the circumcenter of $\triangle CPQ$, so it satisfies $|OC| = |OP| = |OQ|$.

- Since $|OP| = |OC|$ and $|BP| = |BC|$, both O and B lie on the perpendicular bisector of segment PC. Therefore, $BO \perp PC$.
- Since $|OQ| = |OC|$ and $|DQ| = |DC|$, both O and D lie on the perpendicular bisector of segment QC. Therefore, $DO \perp QC$.

Step 4: Orthocenter of $\triangle ABD$

We utilize the parallel properties of the parallelogram $APCQ$:

1. We established $BO \perp PC$. Since $PC \parallel AD$, it follows that $\mathbf{BO} \perp \mathbf{AD}$. (Line BO contains the altitude from B to AD).

2. We established $DO \perp QC$. Since $QC \parallel AB$, it follows that $\mathbf{DO} \perp \mathbf{AB}$. (Line DO contains the altitude from D to AB).

Since O is the intersection of these two altitudes, O is the **orthocenter** of $\triangle ABD$.

Conclusion

Since O is the orthocenter of $\triangle ABD$, the line segment connecting the third vertex A to O must lie on the third altitude of the triangle. Consequently, this altitude is perpendicular to the opposite side BD.

$$AO \perp BD$$

$\square$

Visualization